

Hanif Ridho Abdillah

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ABOUT ME

A Geophysical Engineering student with a strong interest in subsurface exploration, particularly in seismic data interpretation and Carbon Capture Storage (CCS). I enjoy delving into geophysical data analysis to understand geological structures and assess subsurface resource potential and evaluation. Driven by a passion for learning, perseverance, and a desire to grow, I actively engage in research and organizational activities to broaden my knowledge and gain practical experience in the field of geophysics.

EDUCATION

Sumatra Institute of Technology (ITERA)

- Bachelor's Degree in Geophysical Engineering (2022 – Present)
- GPA: 3.21 / 4.00
- Active in geophysics-related organizations, research, and competitions.

Public Senior High School 1 Kepahiang

- Period: 2019 – 2022
- Active in the Young Scientific Research Club.
- Developed skills in photography, videography, and singing through artistic activities.

RESEARCH EXPERIENCE

Caprock Integrity Evaluation for Geological CO₂ Storage, Central Sumatra Basin

Final Research Project - 2026 – *Present*

- Conducted geophysical and geomechanical analysis of the Telisa Formation as a potential caprock for Carbon Capture and Storage (CCS).
- Performed well log interpretation using gamma ray, density, sonic, and elastic property analysis.
- Evaluated brittleness index, acoustic impedance, Poisson's ratio, Young's modulus, and unconfined compressive strength (UCS).
- Integrated petrophysical, rock physics, and geomechanical approaches to assess caprock sealing integrity and mechanical behavior.
- Developed scientific visualizations, crossplots, and presentation materials for research communication.

Potential CCS Development in Igneous Rock Formations in Sumatra, Indonesia

Independent Research Project (2025 -Present)

- Conducted preliminary research on the potential application of Carbon Capture and Storage (CCS) in igneous rock formations in Indonesia, particularly Sumatra.
- Investigated subsurface geological characteristics and their relevance for long-term CO₂ mineralization and storage potential.
- Reviewed regional geological, geophysical, and decarbonization-related studies to support CCS development strategies in Southeast Asia.

INTERNSHIP EXPERIENCE

Geophysical Intern – LEMIGAS (Research and Development Center for Oil and Gas Technology) *July 2025 – August 2025*

- Interpreted 2D/3D seismic data to identify subsurface geological structures and identify hydrocarbon zone.
- Utilized seismic attributes to enhance the accuracy of structural interpretation.
- Integrated regional geological data and well data to improve interpretation reliability.

ORGANIZATIONAL & COMMITTEE EXPERIENCE

Structural Geology Laboratory Assistant (*February 2025 – Present*)

- Prepare materials and demonstrate the use of tools such as geological compasses and rock hammers.
- Evaluate student lab results and provide academic and technical feedback to enhance understanding of structural geology concepts.
- Teach geological mapping techniques using topographic maps and stereonet.
- Assist in the processing and interpretation of structural data for tectonic analysis and natural resource exploration.

President, SEG Student Chapter ITERA (*December 2024 – December 2025*)

- Led and managed the SEG Student Chapter ITERA consisting of 40 active members.
- Develop academic programs such as workshops, training sessions, and geophysical exploration seminars with more than 1000 participants.
- Facilitate networking and mentoring opportunities for geophysical engineering students.

Head of Competition Division, Scientific League, ITERA (*May 2024 – November 2024*)

- Designed the competition framework and mechanisms.

- Coordinated competition staff and collaborated with all involved parties.
- Worked closely with lecturers, judges, and external partners, including academia and industry, to improve competition quality.
- Successfully organized competitions with more than 700 participants.

Research and Professionalism Division Staff, SEG Student Chapter ITERA (2023 – 2024)

- Developed and conducted research for monthly geophysical trivia content.
- Assisted in organizing geophysics seminars.

Competition Subdivision Staff, Scientific League, ITERA (August 2023 – December 2023)

- Structured competition procedures and coordinated with other divisions.
- Assisted in participant organization and event management.

ACHIEVEMENTS / AWARDS

- Grand Finalist Student Competition Himpunan Ahli Geofisika Indonesia Student Chapters Week 2025 using Paleoscan Software
- Man of The Match Student Competition Himpunan Ahli Geofisika Indonesia Student Chapters Week 2025
- Finalist, Student Creativity Program (PKM) 2025
- Participant, Sanitation Essay Competition, Bandung 2023
- Participant, SDGs LITERA Competition 2022

SKILLS

- Communication & Leadership: Able to work collaboratively and lead projects.
 - Languages: Indonesian (Native), English (Basic – Active).
 - Programming: Python.
 - Geophysical Software: Petrel, HRS, Paleoscan, QGIS, Surfer, ReflexW, RES2DINV, Oasis Montaj.
 - Geophysical Data Analysis: Processing and interpreting geophysical data.
 - Microsoft Office: Word and Excel.
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